Enumerative Combinatorics — problem 5.45

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Let $1 \leq r \leq n$ and consider all subsets of r elements of the set $\{1, 2, \ldots, n\}$. Each of these subsets has a smallest member. Let $F(n, r)$ denote the arithmetic mean of these smallest numbers ; prove that

$$F(n, r) = \frac{n+1}{r+1} \qquad \text{(IMO 1981)}.$$